FDS CASE STUDY REPORT

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Semester: 5

Subject Name: Fundamentals of Data Science

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CASE STUDY 1

ONLINE RETAIL TRANSACTION ANALYSIS AND CLUSTERING

GitHub: https://github.com/swarareddy/online-retail-clustering-case-study

1. Problem Statement

The online retail store has transaction records containing multiple products purchased together, but lacks a systematic analysis of transaction characteristics and groups of similar shopping baskets. This case study explores the structure of the baskets, summarizes product and transaction patterns, represents baskets numerically, and compares four clustering algorithms. The aim is to identify potentially useful patterns while clearly acknowledging the limits of the dataset and methods.

2. Methodology

Data loading and cleaning. The CSV was read without a header because the first line contains product entries. Product names were converted to lowercase and stripped of extra whitespace. Empty cells were treated as missing values, and duplicate product names within a basket were reduced to one occurrence.

Transaction representation. Each row was treated as one shopping basket. A binary transaction-item matrix was constructed: 1 means a product appears in the basket and 0 means it does not. This representation records presence, not quantity or revenue.

Dimensionality reduction. PCA was applied to the binary item matrix. The first two components were used for a 2D visualization. For clustering, a reproducible sample of up to 1,200 transactions was represented by 10 PCA components and standardized. The same sample and feature space were used for all four methods to make the comparison more consistent.

Algorithms and parameters. K-means used four clusters; hierarchical clustering used Ward linkage with four clusters; DBSCAN used eps=2.5 and min_samples=8; spectral clustering used four clusters and a 10-nearest-neighbor affinity graph. These are exploratory settings, not necessarily optimal parameters.

3. Statistical Findings

The cleaned dataset contains 7,501 transactions and 119 distinct products. Each transaction contains between 1 and 20 distinct products. The mean basket size is 3.91 products, the median is 3, and the standard deviation is 2.90. The first and third quartiles are 2 and 5 products, respectively.

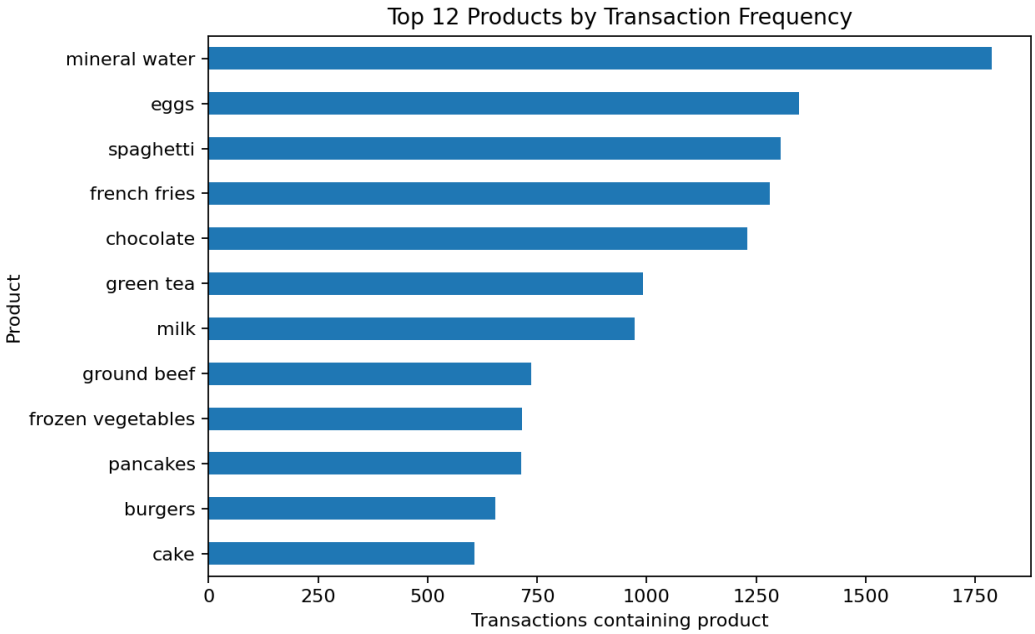


Figure 1. The bars show the number of baskets containing each of the 12 most frequent products. Mineral water is the most frequent item, appearing in 1,788 transactions, followed by eggs (1,348), spaghetti (1,306), French fries (1,282), and chocolate (1,229). Frequencies are basket counts, not units sold.

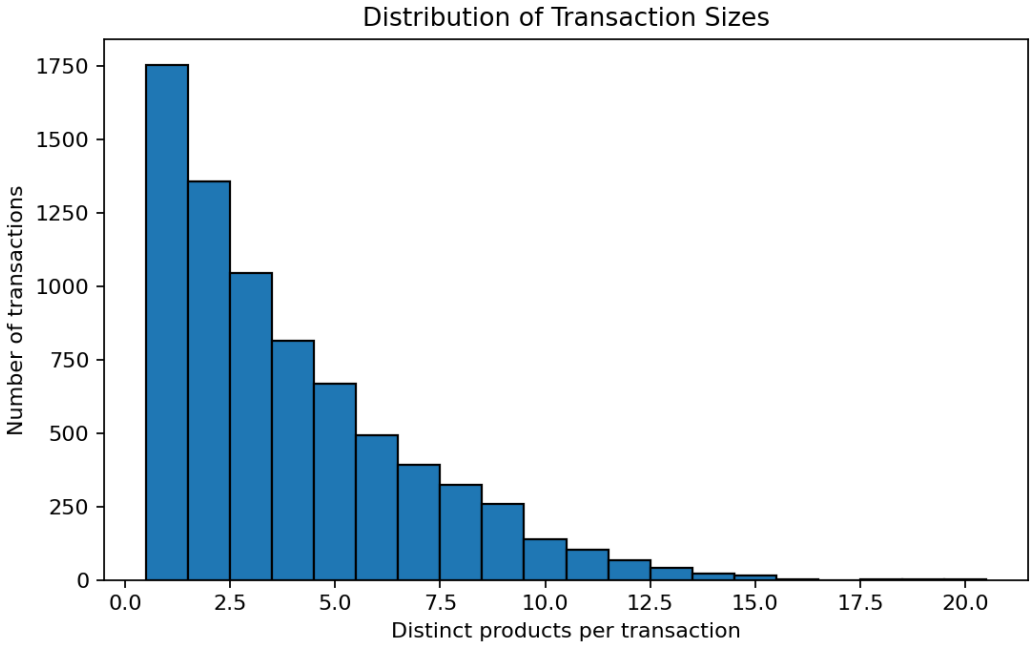


Figure 2. The distribution shows how many transactions contain each number of distinct products. The mean (3.91) is higher than the median (3), indicating that larger baskets pull the average upward. Most baskets are relatively small, while a smaller number contain many products.

4. PCA Findings

The first principal component explains 6.72% of variance and the second explains 4.47%. Together they explain **11.18%** of the variation. Therefore, the 2D projection is useful for a broad visual overview but does not preserve most of the information in the full feature space. Points that appear close in the plot are similar in this projection; apparent overlap does not prove that clustering is impossible.

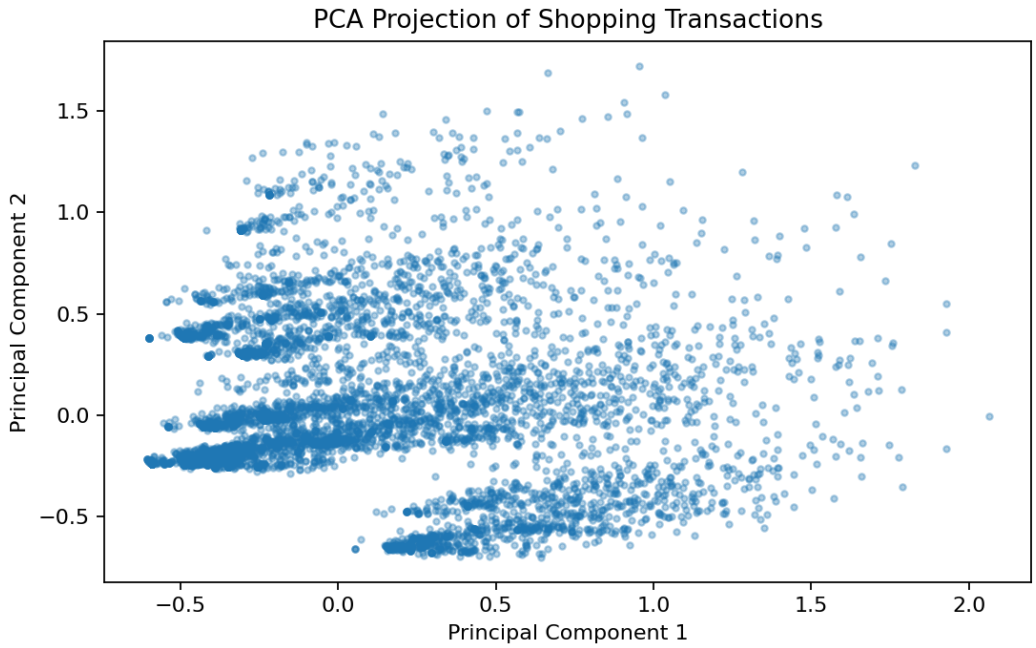


Figure 3. PCA projection of all transaction baskets using the first two principal components.

5. Model Evaluation

Clustering was performed on the same 1,200-transaction sample in a standardized 10-component PCA space. The silhouette coefficient ranges from -1 to 1, with larger values generally preferred. The Davies–Bouldin index is non-negative, with smaller values generally preferred. DBSCAN noise points (label -1) were excluded from its metric calculations, so its scores must be interpreted alongside its noise fraction.

Algorithm	Clusters	Noise fraction	Silhouette ↑	Davies–Bouldin ↓
K-means	4	0.0%	0.248	1.809
Hierarchical	4	0.0%	0.249	1.677
DBSCAN	4	17.9%	0.216	1.227
Spectral	4	0.0%	-0.065	2.453

Interpretation. Hierarchical clustering has the highest silhouette among K-means and hierarchical clustering (about 0.249) and a lower Davies–Bouldin index (1.677) than K-means. DBSCAN has the lowest Davies–Bouldin index (1.227) among these results, but it labels 17.9% of sampled transactions as noise, and its silhouette (0.216) is lower than hierarchical clustering. Spectral clustering has a negative silhouette (-0.065) and the highest Davies–Bouldin index (2.453), suggesting poor separation under these settings. Overall, hierarchical clustering is the strongest balanced result in this particular run; this is not proof it is universally best.

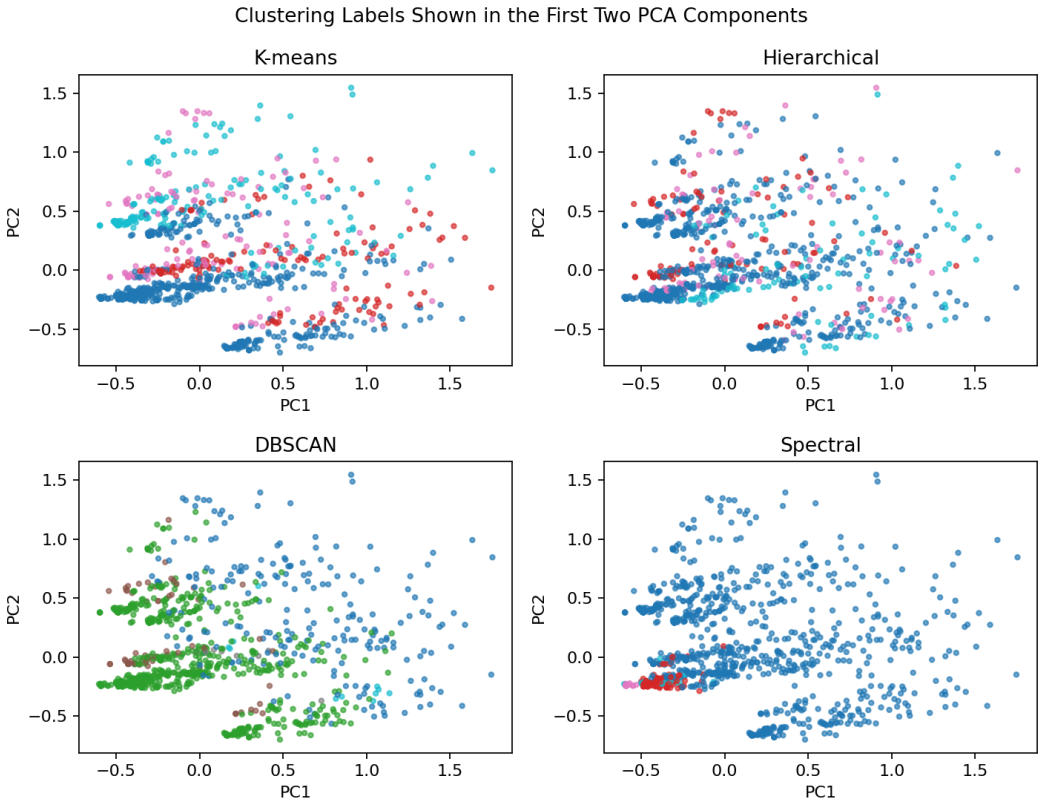


Figure 4. Cluster labels plotted against the first two PCA components. Clustering used 10 standardized PCA components; this figure displays only the first two, so visual overlap may differ from the clustering feature space.

6. Business Implications

- **Product availability:** frequently occurring products such as mineral water, eggs and spaghetti may deserve attention when planning shelf availability, subject to margin and stock data.
- **Cross-selling experiments:** common products within a cluster may suggest combinations worth testing in bundles or recommendations. Association analysis would be more direct for discovering products frequently bought together.
- **Basket-size planning:** the median basket contains three distinct products, which may help inform checkout or promotion experiments, but transaction-level timing and customer context are absent.
- **Pilot before deployment:** validate any proposed promotion against sales, profit, stock and controlled experiments rather than treating cluster membership as causal evidence.

7. Limitations

- The dataset has no customer IDs, dates, prices, quantities or profit information. It clusters baskets, not customers, and cannot support customer lifetime-value conclusions.
- Binary encoding treats every product occurrence equally, regardless of quantity or price.
- The basket matrix is sparse (only about 3.29% of entries are 1), which can make distance-based clustering difficult.
- The 2D PCA plot explains only 11.18% of variance; even 10 PCA components explain only 36.21%. Dimensionality reduction loses information.

- Clustering metrics are calculated on a fixed random sample of 1,200 transactions, not all 7,501 baskets. Results may change with another sample or parameter settings.
- DBSCAN scores exclude noise transactions; its scores must not be interpreted without considering the proportion labelled noise.
- Cluster metrics are mathematical summaries and do not guarantee that groups are commercially meaningful.

8. Conclusion

The analysis of 7,501 transactions and 119 products found an average basket size of 3.91 and a median of 3. Mineral water, eggs, spaghetti, French fries and chocolate were the most frequent products. The first two PCA components explained 11.18% of variance, so the 2D plot provides only a limited view. In the clustering comparison, hierarchical clustering gave the strongest balanced result under the chosen settings, while DBSCAN found a dense majority group and treated a portion of baskets as noise. Spectral clustering performed poorly by the selected metrics in this run. The findings can guide hypotheses for product placement and cross-selling, but should be validated with richer data and business experiments.

CASE STUDY 4

PREDICTING STUDENT DROPOUT AND ACADEMIC SUCCESS

GitHub: https://github.com/swarareddy/Student-Dropout-and-Academic-Success-Prediction

1. Problem Statement and Objectives

The institution wants to understand factors associated with student dropout and academic success, and predict the eventual outcome Dropout, Enrolled or Graduate. Objectives include data-quality checks, descriptive statistics, visualizations, hypothesis testing, multiclass classification, model interpretation and responsible recommendations.

2. Methodology

The semicolon-separated dataset contains 4,424 records and 37 columns. *Target* is the outcome variable. Integer-coded attributes such as course, application mode and qualification are categories, not continuous measurements. Missing values, duplicate rows, ranges and target labels were checked. Numerical fields are median-imputed and standardized; coded categorical fields are mode-imputed and one-hot encoded.

Leakage prevention: the primary model predicts at enrollment, so all 12 first- and second-semester curricular-unit variables were excluded from its predictors. They are examined separately in EDA and hypothesis testing. A stratified 75/25 train-test split and class-weighted logistic regression were used.

3. Statistical Findings

The target classes are Graduate: 2,209 (49.93%), Dropout: 1,421 (32.12%), and Enrolled: 794 (17.95%). Graduate is the largest class and Enrolled is the smallest, so accuracy alone is insufficient.

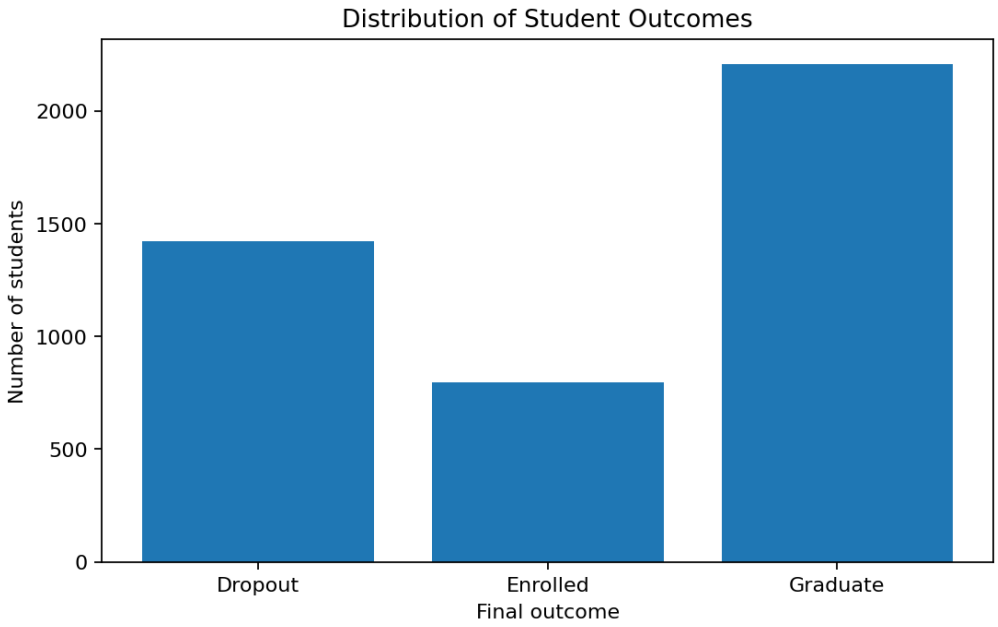


Figure 1. Counts of Dropout, Enrolled and Graduate outcomes. The class distribution is imbalanced.

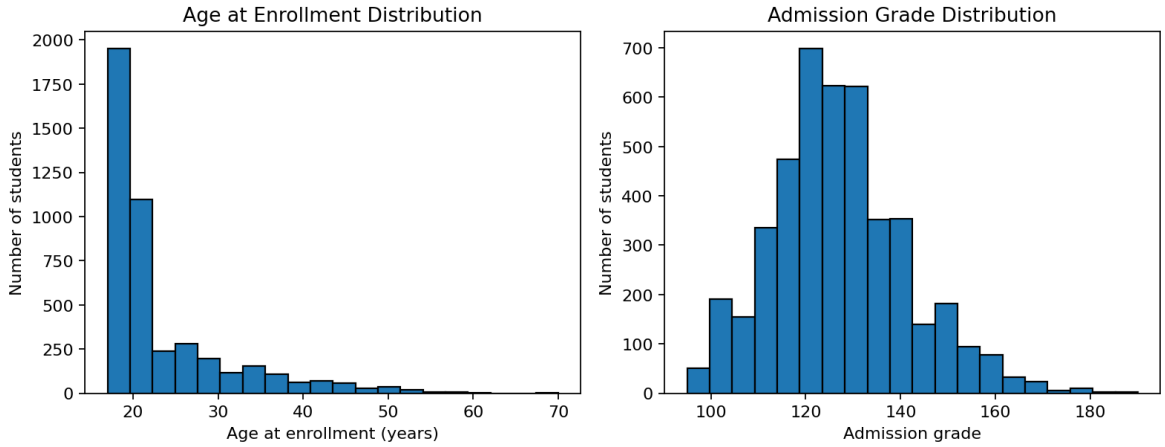


Figure 2. Histograms of age at enrollment and admission grade. Age ranges from 17 to 70; distributions show variation and a right tail in age.

Mean admission grades by outcome are Dropout 124.96, Enrolled 125.53, and Graduate 128.79. The Graduate group has the highest mean, though the groups overlap.

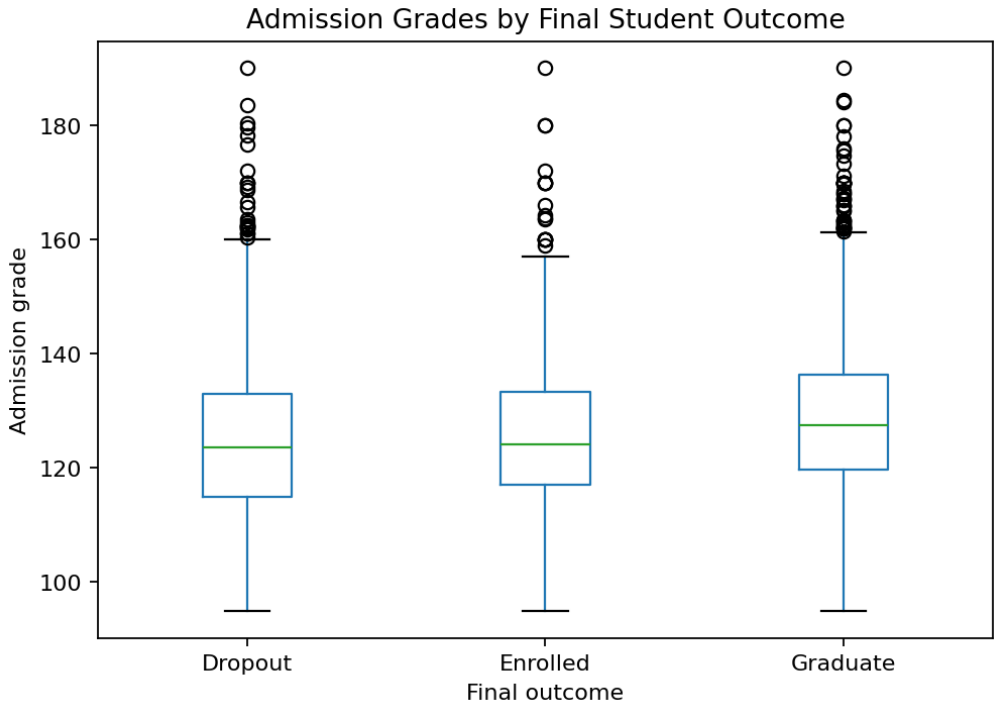


Figure 3. Admission grades by final outcome. Compare medians, spread and overlap; this association does not establish causality.

Mean first-semester grades are Dropout 7.26, Enrolled 11.13, and Graduate 12.64. The dropout group has a notably lower average.

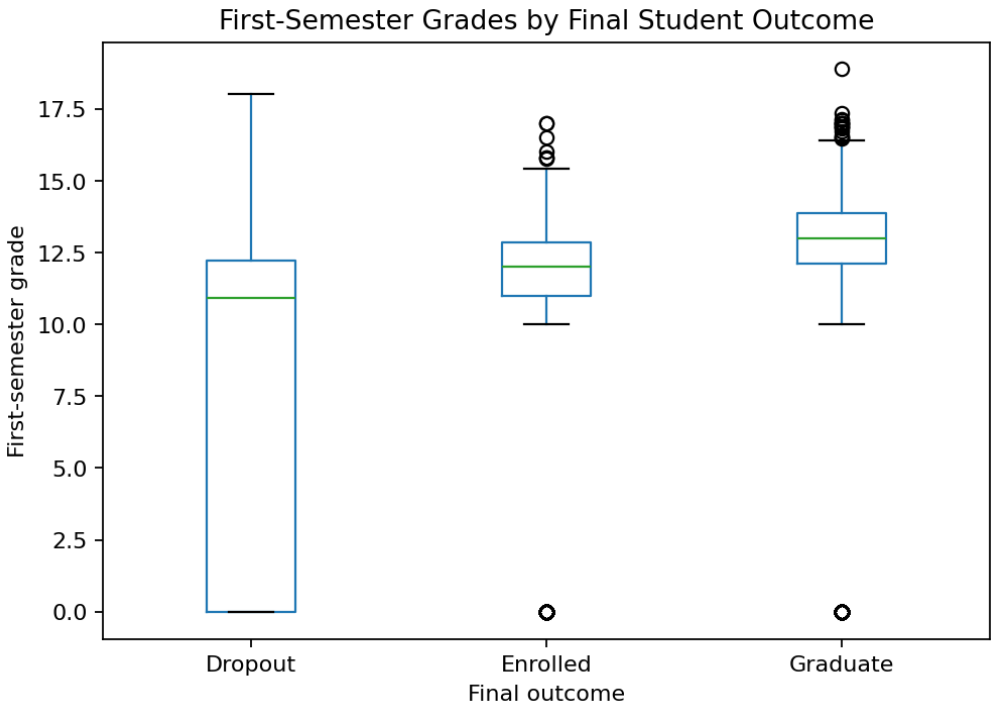


Figure 4. First-semester grade distributions by eventual outcome. These grades may support semester-one monitoring but are excluded from the enrollment-time model.

4. Age and Outcome Composition

Outcome composition differs across age groups. The dropout share is higher in older age bands than among students aged 20 or younger. This suggests a topic for further investigation, not a reason to assume age causes dropout.

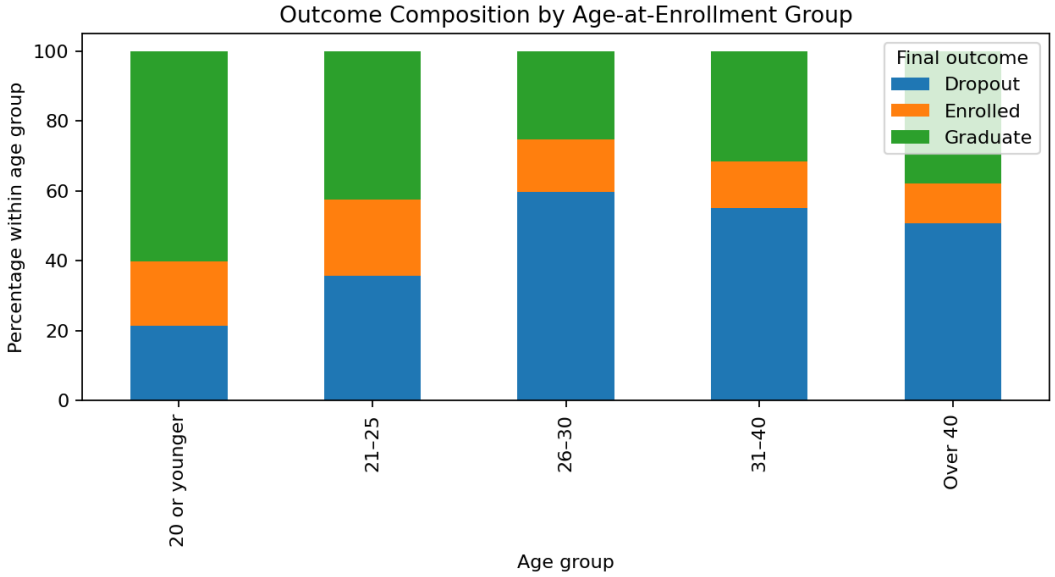


Figure 5. Stacked bars show the percentage of outcomes within each age group; each bar sums to 100%.

5. Hypothesis Test

H_0: first-semester grade distributions are the same across Dropout, Enrolled and Graduate. H_1: at least one distribution differs. The Kruskal–Wallis test compares three independent groups without requiring normality; observations should be independent, and very different distribution shapes limit median-only interpretations.

At α = 0.05, Kruskal–Wallis H = 1094.186 and p = 2.51e-238. Reject H_0: there is strong statistical evidence of differences in first-semester grade distributions. Approximate epsilon-squared = 0.247, indicating a non-trivial group association in this sample. Statistical significance does not prove causation.

6. Model Evaluation

On the held-out test set of 1,106 students, accuracy = 0.575; macro precision = 0.561, macro recall = 0.560, and macro F1 = 0.549. Macro metrics give each class equal weight.

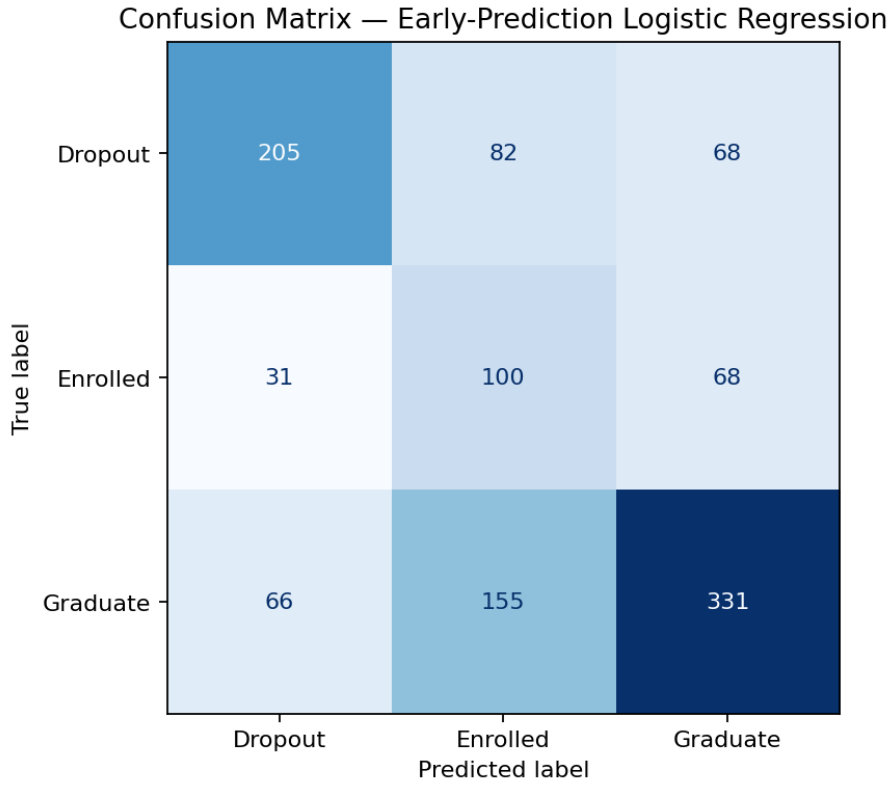


Figure 6. Confusion matrix. Rows are actual outcomes and columns are predicted outcomes; off-diagonal cells represent classification errors.

Class	Precision	Recall	F1-score	Support
Dropout	0.679	0.577	0.624	355
Enrolled	0.297	0.503	0.373	199
Graduate	0.709	0.600	0.650	552
Macro avg	0.561	0.560	0.549	1106

F1 is higher for Graduate (0.650) and Dropout (0.624) than for Enrolled (0.373). Enrolled precision is only 0.297, so many predictions of Enrolled are incorrect. Class weighting improves attention to

smaller classes but does not eliminate errors.

7. Model-Derived Feature Interpretation

Dropout: largest positive class coefficients include Mother's occupation_0 (1.50), Father's occupation_90 (1.05), Mother's occupation_90 (1.04), Mother's qualification_9 (0.86). Coefficients represent conditional associations in the fitted model; coded-category coefficients depend on reference coding and are not causal effects.

Enrolled: largest positive class coefficients include Father's qualification_39 (1.11), Course_9119 (0.89), Mother's occupation_192 (0.88), Application mode_5 (0.83). Coefficients represent conditional associations in the fitted model; coded-category coefficients depend on reference coding and are not causal effects.

Graduate: largest positive class coefficients include Application mode_15 (1.35), Marital status_5 (0.87), Nacionality_11 (0.78), Father's qualification_43 (0.64). Coefficients represent conditional associations in the fitted model; coded-category coefficients depend on reference coding and are not causal effects.

8. Limitations

- The model predicts recorded historical outcomes and may not generalize to another institution or future cohort.
- The early model excludes semester performance to avoid leakage, so it may perform less well than a model built after semester one.
- Class imbalance and low Enrolled precision mean class-specific errors require attention.
- Logistic regression may miss nonlinear relationships and complex interactions.
- Demographic associations can reflect structural inequities and must not be used to label or penalize individuals.
- The target is status at the end of normal course duration, not necessarily eventual long-term completion.

9. Recommendations

- Offer voluntary tutoring, advising, academic planning and financial-aid guidance to students who may benefit.
- Use first-semester performance for a separately validated early-warning process and timely supportive outreach.
- Investigate scheduling, work, family, financial and resource barriers rather than attributing dropout to demographic traits.
- Audit errors and performance across relevant student groups, provide human review and allow records to be corrected.
- Evaluate support interventions prospectively to learn whether they improve outcomes.

10. Conclusion

The dataset has 4,424 students and 37 columns. Graduate is the largest class (49.93%) and Enrolled the smallest (17.95%). First-semester grade distributions differ significantly ($H = 1094.186$, $p < 0.001$). The early-prediction logistic regression achieved 0.575 accuracy and 0.549 macro F1, with weaker performance on Enrolled. The model may guide supportive outreach and further investigation, but it is not causal evidence and should not be used for punitive decisions.

Submission note: keep Studentdata.csv beside the notebook, run all cells, fill in the title page and verify that you understand the results and methods. Follow your faculty's attribution rules.